

Representation Theory in Quantum Physics: The Mustard Behind the Magic

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1 Motivating Examples: The Symmetric Group

Consider particles 1 and 2 with positions x_1 and x_2 . S_2 acts on this system and so it gives rise to a representation where we can take our vector space of states to be

$$V = \mathbb{C}\{\psi(1,2), \psi(2,1)\}$$

where $\psi(i,j)$ denotes particle i at x_1 and particle j at x_2 . If we just let S_2 act freely on V then

$$e \leftrightarrow \begin{matrix} \psi(1,2) & \psi(2,1) \\ \psi(1,2) & \left(\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right) \\ \psi(2,1) & \end{matrix}, \quad (1\ 2) \leftrightarrow \begin{matrix} \psi(1,2) & \psi(2,1) \\ \psi(1,2) & \left(\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right) \\ \psi(2,1) & \end{matrix}$$

meaning that $\chi = (2,0) = \chi_{trv} + \chi_{alt}$. What do these irreps correspond to? We find an eigenbasis of the permutation $(1\ 2)$:

$$\begin{aligned} \vec{v}_1 &= \psi(1,2) + \psi(2,1) \implies (1\ 2)\vec{v}_1 = \vec{v}_1 \implies \chi_+ = (1,1) \\ \vec{v}_2 &= \psi(1,2) - \psi(2,1) \implies (1\ 2)\vec{v}_2 = -\vec{v}_2 \implies \chi_- = (1,-1) \end{aligned}$$

We can do a similar construction with 3 particles with positions $\{x_1, x_2, x_3\}$ where $\psi(i,j,k)$ means that particle i is in position x_1 and so on.

$$\begin{aligned} V_{trv} &\leftrightarrow \mathbb{C}\vec{v} \text{ s.t. } \vec{v} = \sum_{(i,j,k) \in P_3} \psi(i,j,k) \\ V_{alt} &\leftrightarrow \mathbb{C}\vec{v} \text{ s.t. } \vec{v} = [\psi(1,2,3) + \psi(3,1,2) + \psi(2,3,1)] - [\psi(1,3,2) + \psi(2,1,3) + \psi(3,2,1)] \\ V_{std} &\leftrightarrow \mathbb{C}\{\vec{v}_1, \vec{v}_2\} \\ \vec{v}_1 &= [\psi(1,2,3) + \psi(3,1,2) + \psi(2,1,3)] - [\psi(2,3,1) + \psi(1,3,2) + \psi(3,2,1)] \\ \vec{v}_2 &= [\psi(1,2,3) + \psi(3,2,1) + \psi(2,1,3)] - [\psi(2,3,1) + \psi(1,3,2) + \psi(3,1,2)] \end{aligned}$$

Can check that these vector spaces are irreducible and closed under S_3 action, moreover the characters correspond, as an example, for the standard rep we have

$$\begin{aligned} (1\ 2) \circ \vec{v}_1 &= [\psi(1,2,3) + \psi(3,2,1) + \psi(2,1,3)] - [\psi(2,3,1) + \psi(1,3,2) + \psi(3,1,2)] = \vec{v}_2 \\ (1\ 2\ 3) \circ \vec{v}_1 &= [\psi(2,3,1) + \psi(3,1,2) + \psi(1,3,2)] - [\psi(1,2,3) + \psi(3,2,1) + \psi(2,1,3)] = \vec{v}_2 \end{aligned}$$

and similar for other actions.

2 Why we care

1. The physical act of measuring an N -particle system forces our ambiguous quantum state to “collapse” to one of the eigenstates of the operator corresponding to our observable (e.g. position, momentum, spin).
2. But as the above examples have shown, these eigenstates themselves are ambiguous, as for two bosons, the particles occupy the same spin state, while for two fermions the particles occupy different spin states, yet we cannot tell which particle is which among the two.
3. That being said the S_n action gives us coarser information about what states the collection of identical particles as a whole can be in.

Remark : though signs come into play when expressing the eigenvectors as linear combinations of the state vectors, the state vectors can be taken to be orthogonal, so that when we calculate inner products (which is how we calculate probabilities of what state the system is in), the signs are irrelevant.

3 Background

Def : The “state space” of a quantum system is a f.d. complex vector space with a (Hermitian) inner product, where the elements are called wave functions. An element of the state space is denoted by ψ or $\psi(x, t)$

Def : An observable is a self-adjoint linear operator on the state space.

$$\hat{O} : V \rightarrow V \text{ s.t. } \hat{O}^t := (\hat{O}^*)^T = \hat{O}$$

- When observables do not commute, physics becomes interesting.
- Many different observables can act on one wave function to garner different information, e.g.

$$\hat{X} \rightarrow x, \quad \hat{P} \rightarrow -i\hbar \frac{\partial}{\partial x}, \quad \hat{K}E \rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$$

- Observable states of an operator $\hat{O}$ are just the eigenstates, but called such because they can be thought of as states for which the observable has a well defined value.

4 Representations

Def : A Lie Group is a group with the elements forming a smooth manifold with group multiplication being smooth (i.e. infinitely differentiable). We will assume that given a f.d. representation on a Lie group, that said representation is smooth.

Example: $\mathbb{R}$ under addition, or $SO(2) \cong S^1 \cong U(1)$ under addition of angle/multiplication of complex numbers.

Def A representation, (π, V) , for V a f.d. v.s. over $\mathbb{C}$ with Hermitian inner product is unitary if

$$\forall g \in G, \quad \langle \pi(g)\vec{v}_1, \pi(g)\vec{v}_2 \rangle$$

In matrix form, this means that for $M \leftrightarrow \pi(g)$ we get $M^{-1} = M^t$.

Remark : Any time we have a group acting on a quantum system, we get a unitary representation. In particular lie group actions give rise to observables on state spaces.

Example: Consider $\mathbb{R}$ again and the schrodinger equation:

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \hat{H} \psi(x, t) \implies \psi(x, t) = e^{-(i/\hbar)\hat{H}t} \psi(x, 0)$$

this solution makes sense in some contexts, and assuming it does, it gives rise to a 1 dimensional representation $(\pi, \mathbb{R})$ such that $\pi(t) = e^{-(i/\hbar)Ht}$, which is clearly a homomorphism. For future reference, we note that the the Schrodinger equation dictates how our state space transforms in time and $\hat{H}$ is called the Hamiltonian, and the operator $\widehat{U}(t) := e^{-(i/\hbar)Ht}$ is called the propagator, which can be thought of a flow. Moreover

$$\langle \pi(t)\psi(x, t_1), \pi(t)\psi(y, t_2) \rangle = \overline{\pi(t)\psi(x, t_1)} \pi(t)\psi(y, t_2) = [\overline{\pi(t)}\pi(t)][\overline{\psi(x, t_1)}\psi(y, t_2)] = \langle \psi(x, t_1), \psi(y, t_2) \rangle$$

Theorem 4.1. Any unitary representation is completely reducible, i.e. can be written as a direct product of irreducible representations.

Proof: The standard proof involves constructing a G -invariant Hermitian inner product, but by definition of unitary representation, we’re given one and can proceed as follows: if there exist $W \subseteq V$ is preserved under G -action then $Z = W^\perp$ is closed under G -action as

$$\forall z \in Z, \quad \forall w \in W, \quad \langle \pi(g)z, w \rangle = \langle \pi(g)z, \pi(g)\pi(g^{-1})w \rangle = \langle z, \pi(g^{-1})w \rangle = 0$$

as $\pi(g^{-1})w \in W$, so $V = W \oplus Z$ and we can continue forward. $\square$

Theorem 4.2. If a group G is commutative, all of its irreps are one-dimensional

Proof: Consequence of Schur's lemma, which holds in the infinite dimensional case. (There might be extra conditions on the group in the case that it is infinite)

Theorem 4.3. *All irreps of $U(1)$ are unitary and given by*

$$\pi_k(e^{i\theta}) = e^{ik\theta}, \quad k \in \mathbb{Z}$$

Proof: For any representation π , we write $\pi(\theta)$ to mean $\pi(e^{i\theta})$ (lifting to $\mathbb{R}$ as a cover), then we have two conditions to impose $\pi(\theta_1 + \theta_2) = \pi(\theta_1)\pi(\theta_2)$ and $\pi(2\pi n) = \pi(0) = 1$. From the homomorphism property, we get that

$$\pi(\theta) = Ce^{\alpha\theta}, \quad \pi(0) = 1 = \pi(2\pi n) \implies C = 1, \quad \alpha = ik, \quad k \in \mathbb{Z}$$

so all irreps are π_k .

5 Conservation and Commutation

Recall: our example of the Schrodinger equation and also the linear algebra fact that commuting matrices (which are themselves diagonalizable) are simultaneously diagonalizable. Then we're interested in operators, $\hat{O}$, which commute with the Hamiltonian, i.e. $[\hat{H}, \hat{O}] = 0$, as

$$[\hat{O}, \hat{H}] \implies [\hat{O}, \widehat{U(t)}] = 0$$

In particular, if $\psi(x, 0)$ is an observable/eigenstate with value λ , then

$$\hat{O}\psi(x, t) = \hat{O}\widehat{U(t)}\psi(x, 0) = \widehat{U(t)}\hat{O}\psi(x, 0) = \lambda\widehat{U(t)}\psi(x, 0) = \lambda\psi(x, t)$$

i.e. the value corresponding to $\hat{O}$ is conserved over time.

Finding operators which commute with the Hamiltonian and then identifying further commutation relations is the back bone of basic quantum physics.

Example: (see Shankar p. 321). The angular momentum of individual particles is important in Physics and for *rotationally invariant systems*, we construct angular momentum operators, L_x, L_y, L_z , so that $[H, L_i] = 0$ and hence for $L^2 = L_x^2 + L_y^2 + L_z^2$, $[H, L^2] = 0$. From this, we learn that angular momentum is conserved over time, and that energy does not depend on the orientation of the system. From these operators, we also get unitary representations

$$\widehat{O}_z(\theta) = \exp(-i\theta L_z/\hbar)$$

by our work on Unitary representations, we see that $\widehat{O}_z(\theta) = \widehat{O}_z(e^{i\theta}) = e^{ik\theta}$ for some $k \in \mathbb{Z}$. Turning to an eigenbasis of L_z , the irreps of $\widehat{O}_z$ are one-dimensional and correspond to the eigenspaces of L_z . This implies that L_z must have integer eigenvalues, from which we can label the eigenspaces of L_z . More importantly, this implies that for particles in rotationally invariant systems, angular momentum is discrete (in a given component), which is shockingly different from classical physics (which I will now demonstrate).